

Topology Optimization of the Hub Carrier of an ultra-efficient lightweight electric vehicle

- TOPOLOGY OPTIMISATION

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Introduction

The subject of the project is an ultra-efficient lightweight electric vehicle (Figure 1) designed and manufactured by the Shell Eco-marathon team of Politecnico di Milano.



Figure 1: the ultra-efficient vehicle designed to compete in the Shell Eco Marathon.

The main goal that is followed during the design of this type of vehicle is obviously the maximization of energy efficiency while guaranteeing a proper vehicle handling. The main mechanisms of energy dissipation/usage in a road vehicle are: its mass ($\int_{x_0}^{x_f} a \cdot M_{tot} dx$), tire rolling resistance and aerodynamic drag. An analysis conducted by the student team of Politecnico di Milano showed that mass is the greater contributor to energy consumption for this type of vehicle (Figure 2).

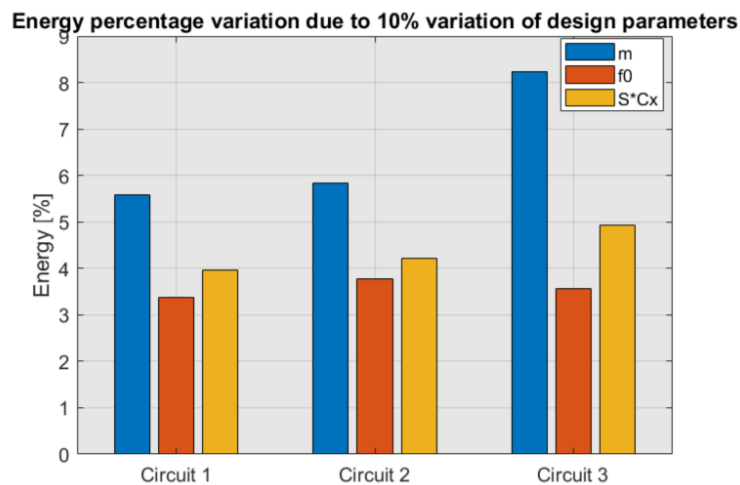


Figure 2: sensitivity analysis on energy consumption for three different circuits.

As mass is the main contributor to energy consumption, it's extremely important to optimize with respect to mass the components to manufacture, while guaranteeing the required stiffness.

For this reason, the aim of this project is to optimize the structural layout of the hub carrier to maximize the structural stiffness with the least amount of mass. The new car is equipped with a double-wishbone suspension, as the one shown in Figure 3, where the hub carrier serves three functions: links the two control arms, converts the force given by the steering rod into a torque used to steer the wheel and it sustains the dynamic and static loads coming from both the wheel, in contact with the asphalt, and the brake calliper.

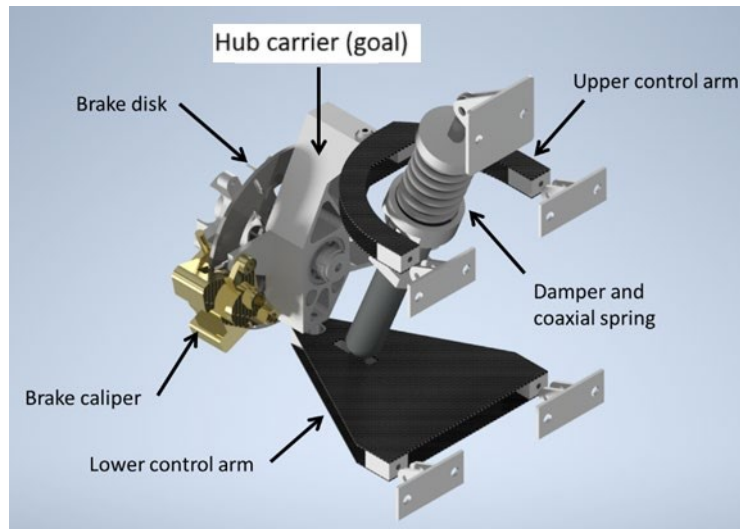


Figure 3: front-right double-wishbone suspension of the vehicle.

First, the design domain of the hub carrier (Figure 4) was defined being aware of the space available around the suspension region and the holes needed to constrain it with the other components.

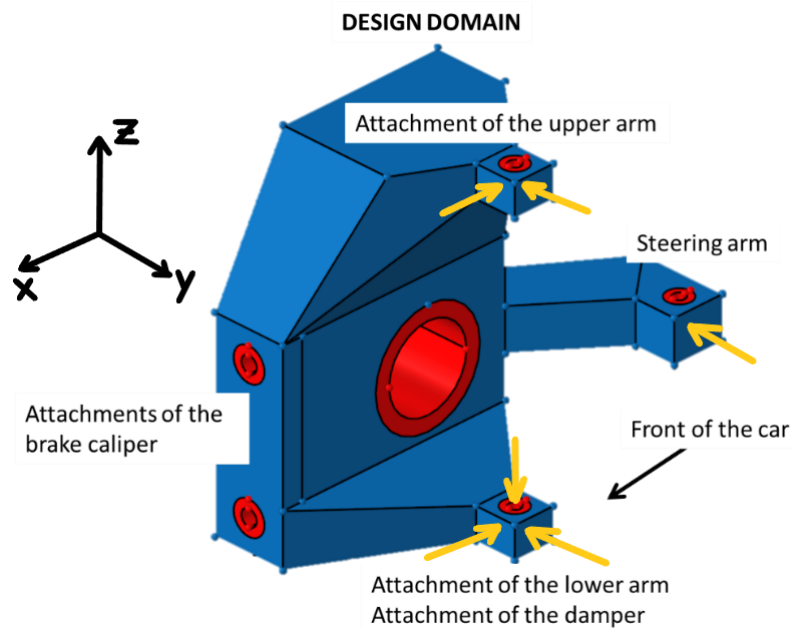


Figure 4: design domain (blue volume) and non-design domain to be conserved (red volume). With yellow arrows are represented the constraints applied on the component.

Because the lower attachment is linked to both lower arm and damper, it was decided to constrain it with a 3D pin. The upper attachment was constrained with a 3D roller (free vertical displacement), being only connected to the upper arm of the double-wishbone suspension. Finally, the steering attachment point was constrained only in the direction along the steering rod translation.

The topology optimization considers all the main loading conditions, summarized in the list below:

- Pure rolling (coasting), considering 160 kg as the mass of the vehicle (Figure 5 - left).
- Braking (Figure 5 - right).
- Internal turn (Figure 6 - left).
- External turn (Figure 6 – right).
- Internal obstacle (Figure 7 - left).
- External obstacle (Figure 7 - right).

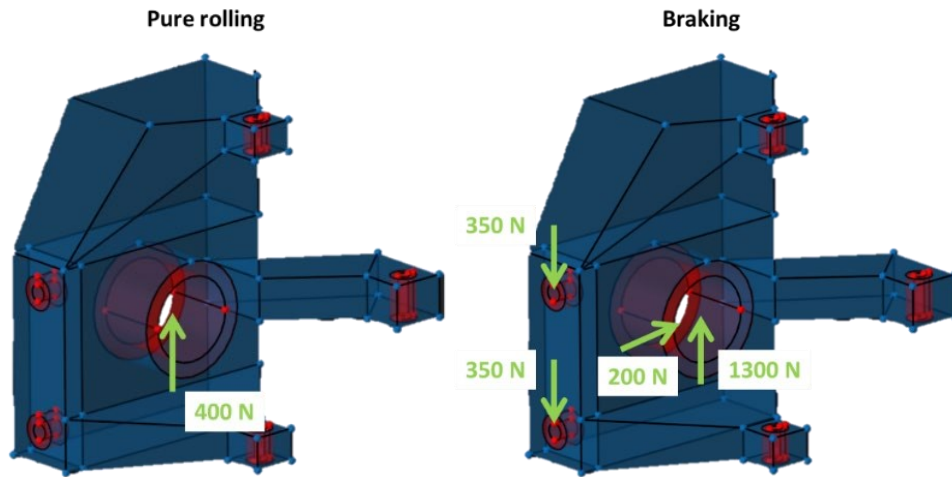


Figure 5: schematics of the loads applied in pure rolling (left) and braking (right).

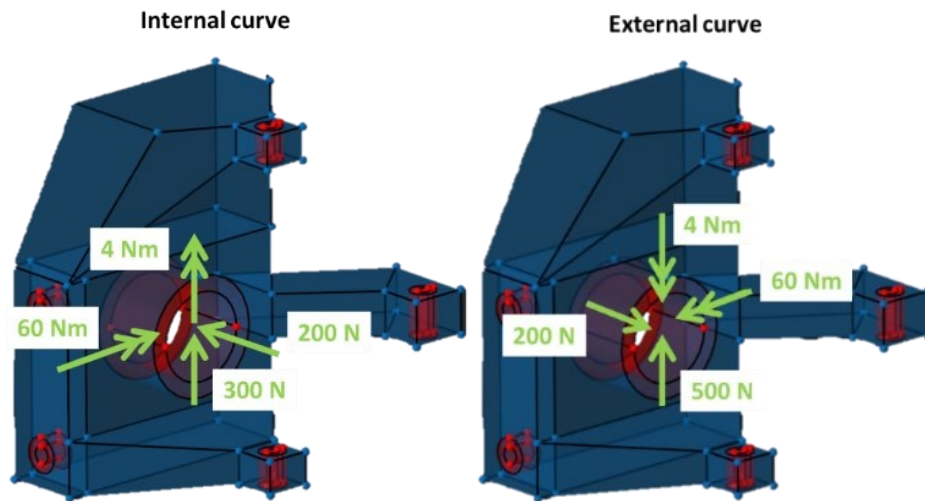


Figure 6: schematics of the loads applied when taking a curve with the hub carrier on the internal side (left) and external one (right).

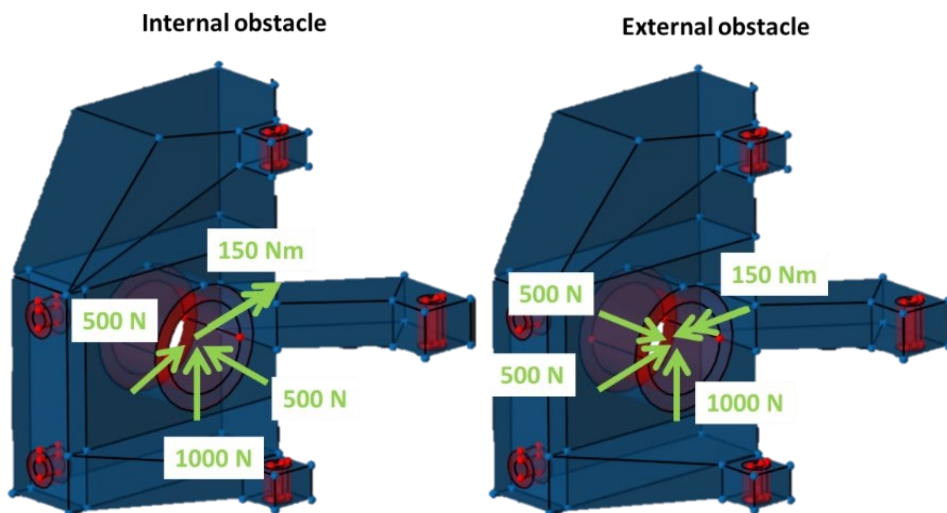


Figure 7: schematics of the loads applied on the hub carrier when the vehicle rolls due to the presence of an obstacle on one side or the other with respect to this component.

The material used to manufacture the hub carrier is the Aluminium alloy 6061 T6 ($E=70000$ MPa, $\rho = 2700$ kg/m³, $\sigma_Y = 270$ MPa).

The objective function of the topology optimization problem is the weighted compliance, that is the sum of the compliance given by each loading condition, multiplied by a weight, to set the level of contribution to the load case. The behavioural constraints are imposed as camber stiffness, toe stiffness and mass fraction. The camber stiffness is measured from the centre of the hub carrier when a rotation around the x-axis is applied, while the toe stiffness is measured in the same way except for the axis used, in this case the z one. The optimization problem is summarized as follows:

$$\left\{ \begin{array}{l} \min_{\rho^*} \sum_{i=1}^{6(\text{load cases})} \omega_i \bar{F}_i^T \bar{u}_i \\ \text{camber stiffness} \geq 160000 \text{ Nm/rad} \\ \text{toe stiffness} \geq 84000 \text{ Nm/rad} \\ \frac{m_{opt}}{m_{tot}} < \text{massFrac}_{max} (< 1) \leftrightarrow m_{opt} - \text{massFrac}_{max} * m_{tot} \leq 0 \end{array} \right.$$

A forging (along y-direction) manufacturing constraint was chosen to obtain the optimized hub carrier with a manufacturable geometry.

Density-based topology optimization algorithm

Altair OptiStruct, the solver used in this project, applies a density-based topology optimization algorithm usually incorporating the SIMP (Solid Isotropic Material with Penalization) method. This approach links a design variable ρ_i^* to the stiffness matrix of element i and promotes pseudo-density values close to either one (solid material) or zero (void) through penalization.

The algorithm main steps are: solving the static structure with FEM for all the load cases, calculating separately and summing all the compliances with weights, computing the weighted compliance sensitivities with respect to the densities and with respect to the intervening design variables if a manufacturing constraint is applied, updating these variables based on the optimization criteria, repeating these steps until convergence.

The objective function of this problem can be proved to be convex, as the behavioural constraints (linear), so the overall problem is also convex. As the minima of the problem is clearly constrained, thanks to the convexity of the problem, the Karush-Kuhn-Tucker (KKT) conditions when satisfied by the point $(\bar{\rho}^*, \bar{\lambda})$ will guarantee a global minimum. Due to the non-linearity of the KKT conditions, the Lagrangian duality (minimization and maximization problem) is typically exploited to satisfy these conditions, but before that, the objective function and the behavioural constraints are linearly approximated. The mass fraction constraint as well as the camber and toe stiffness constraints are already linear, while the objective function (weighted compliance) is linearized with intervening variables, probably following the optimality criteria (OC) method:

$$C(\bar{\rho}) \approx C(\bar{\rho}^k) + \sum_{i=1}^{\# \text{ elements}} \frac{\partial C}{\partial y_i} \Big|_{\bar{\rho}=\bar{\rho}^k} (y_i - y_i^k)$$

Where C is the compliance of one of the six load cases and y_i is the intervening variable. As it can be proved that for this static problem the sensitivities of the compliance are always negative, the y_i variables follow a reciprocal linearization:

$$y_i = \frac{1}{\rho_i^\alpha}$$

The $\frac{\partial C}{\partial y_i}$ term can be evaluated as:

$$\frac{\partial C}{\partial y_i} = \frac{\partial C}{\partial \rho_i} \frac{\partial \rho_i}{\partial y_i}$$

Where $\frac{\partial \rho_i}{\partial y_i}$ becomes:

$$\left(\frac{\partial \rho_i}{\partial y_i}\right)^k = \frac{1}{\frac{\partial y_i}{\partial \rho_i}} = \frac{1}{\frac{\partial (\rho_i^{-\alpha})}{\partial \rho_i}} = -(\rho_i^k)^{\alpha+1}$$

While the sensitivities $\frac{\partial C}{\partial \rho_i}$ can be written as:

$$\frac{\partial C}{\partial \rho_i} = \frac{\partial (\bar{F}^T \bar{u}(\bar{\rho}))}{\partial \rho_i} = \frac{\partial (\bar{u}^T [K] \bar{u}(\bar{\rho}))}{\partial \rho_i} = \bar{u}^T [K] \frac{\partial \bar{u}(\bar{\rho})}{\partial \rho_i}$$

Being $\bar{u}$ the nodal displacement vector, $\bar{F}$ the nodal force vector (independent from $\bar{\rho}$).

Where the global stiffness matrix $[K]$ is defined following the Solid Isotropic Material Penalization (SIMP) method:

$$[K](\bar{\rho}^k) = \sum_{i=1}^{\# elements} \int_{V_{el}} [B_i]^T [D] [B_i] = \sum_{i=1}^{\# elements} (\rho_i^k)^p \int_{V_{el}} [B_i]^T [D_0] [B_i] = \sum_{i=1}^{\# elements} (\rho_i^k)^p [k_i^0]$$

Where $[D_0]$ is the material stiffness matrix, which is multiplied by $(\rho_i^k)^p$ as the SIMP penalization method is used to avoid intermediate normalized densities (Young's modulus $E(\rho) = E_{mat} \rho^p$).

So, the sensitivities $\frac{\partial \bar{u}(\bar{\rho})}{\partial \rho_i}$ can be computed by following concept:

$$[K](\bar{\rho}) \bar{u}(\bar{\rho}) = \bar{F}$$

Differentiating it:

$$\frac{\partial [K]}{\partial \rho_i} \bar{u} + \frac{\partial \bar{u}}{\partial \rho_i} [K] = \bar{0} \leftrightarrow \frac{\partial \bar{u}}{\partial \rho_i} = -[K]^{-1} \frac{\partial [K]}{\partial \rho_i} \bar{u}(\bar{\rho}^k) = -[K]^{-1} p(\rho_i^k)^{p-1} [k_i^0] \bar{u}(\bar{\rho}^k)$$

Finally getting the definition of the sensitivities $\frac{\partial C}{\partial \rho_i}$ at iteration k:

$$\left(\frac{\partial C}{\partial \rho_i}\right)^k = -\bar{u}^T [K] [K]^{-1} p(\rho_i^k)^{p-1} [k_i^0] \bar{u} = -p(\rho_i^k)^{p-1} \bar{u}^T [k_i^0] \bar{u}$$

From which the Lagrangian can be written as (for $\alpha = 1$, CONLIN approximation method):

$$L(\bar{\rho}, \bar{\lambda}) = C(\bar{\rho}^k) - \sum_i p(\rho_i^k)^p \bar{u}^T [k_i^0] \bar{u} + \sum_i p(\rho_i^k)^{p+1} \bar{u}^T [k_i^0] \bar{u} \rho_i^{-1} + \lambda_1 (\bar{\rho}^T \bar{v}_{el} - V_0) + \lambda_2 (160000 - k_{camber}) + \lambda_3 (84000 - k_{toe})$$

By exploiting the Lagrangian duality it is possible to get the OC update rule for this topology optimization problem. This update rule being valid only if ρ_i^{k+1} is between ρ_{min} (>0 to avoid numerical singularities) and ρ_{max} set by the user, otherwise ρ_i^{k+1} is set to be equal to the design constraint not satisfied.

The parameters α of the OC method and p of the SIMP method are suitably chosen by the software, but they can be modified by the user. The OC method isn't the only optimization algorithm that Altair Optistruct can use/choose.

FE model setup

The software used for the FE modelling/analysis and topology optimization with density-based approach are: Altair HyperMesh for pre-processing, Altair Optistruct as the solver and Altair HyperView for the post-processing. The fundamental units of measurement used in these programs were t (tonne), mm and s, so to use $N = \frac{t \cdot mm}{s^2}$ and Nmm for the force and moment respectively.

The initial model of the hub carrier was already partitioned into two different regions, assigned to two components: design (blue region of Figure 4) and non-design (red regions of Figure 4), considering the necessary coupling zones with brake calliper, lower control arm, upper control arm, steering rod and wheel axle.

A new material (aluminium alloy) was created with linear-elastic properties (MAT1), a longitudinal elastic modulus $E = 70000 \text{ MPa}$, a Poisson coefficient $\nu = 0.3$ and a density $\rho = 2.7 \cdot 10^{-9} \text{ t/mm}^3$.

A new property with solid section and the material just created was assigned to the design component of the model. The same was done for the non-design component with a differently named property.

A third component was created to store the kinematic couplings between the cylindrical inner surface of the non-design regions and the auto defined node at its centre.

A load collector was created to collect all the nodal constraints applied to the component following the arrangement of Figure 4. Six other load collectors were generated to collect the nodal forces and moments of each loading scenario, following Figure 5, Figure 6 and Figure 7. Finally, six static load steps were created for the FE simulation of each loading case.

Due to the limit imposed by the student license in terms of useable nodes for the FE discretization (max. 100000 nodes), it was preferred to use first order tetrahedral elements (TET4) to get a higher resolution of the optimized component, with the disadvantage being the poor description of displacement and strain field given by this type of element, possibly underestimating the magnitude of stress concentration areas. In fact, as an example, the polynomial approximation of the displacement field u (along x natural coordinate) inside a TET4 element is describe as follows (isoparametric formulation):

$$u(\xi, \eta, \zeta) = (1 - \xi - \eta - \zeta)u_1 + \xi u_2 + \eta u_3 + \zeta u_4$$

It is done similarly for the remaining displacement fields. Where ξ, η, ζ are the material or Lagrangian coordinates. So, the displacement field inside the TET4 element is linearly approximated. Consequently, the strain and stress fields are constant within element boundaries, needing more elements to well capture high stress/strain gradients. For this reason, the maximum number of nodes available was used.

The static study of all the loading conditions was conducted (average mesh size 2.1 mm) to find the highest stress gradient regions. The stress field was qualitatively similar in all cases, in Figure 8 is represented the most critical loading condition: internal obstacle. As expected, the highest stress and highest stress gradients are found close to what would be the spherical joint used to connect the hub carrier to the lower arm and the damper. Therefore, the three solids close to this zone were meshed with TET4 elements of average size 1.8 mm, while the remaining solids were meshed with the same type of elements of size 2.2 mm, the total number of nodes used was 93543 (also counting the 6 kinematic couplings) and the number of elements was 503568.

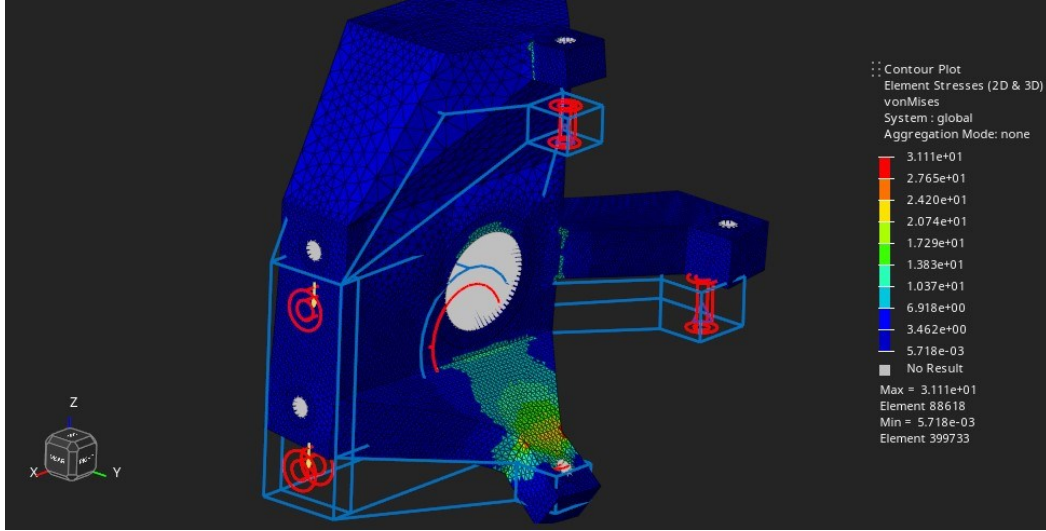


Figure 8: deformed (deformation scale factor = 1000) hub carrier and stress field representation for the internal obstacle case.

Before passing to the setup of the topology optimization problem, two moments (along x and z axis) applied at the centre of the hub were added to two new load collectors to evaluate the camber and toe stiffness, behavioural constraints of the optimization case. More precisely the following values were chosen for their magnitude:

$$M_{x_{camberStiffEval}} = 160000 \text{ Nmm}$$

$$M_{z_{toeStiffEval}} = 84000 \text{ Nmm}$$

Topology optimization: model formulation

First, a new design variable was created by choosing the solid property representing the design component and its material properties, already created at the beginning of the FE modelling phase. The minimum diameter size, which is used by the optimization algorithm filter to avoid checkerboard effects and obtain a manufacturable component (in terms of smallest features), was set to three times the average mesh size ($2 \cdot 3 = 6 \text{ mm}$) that seems reasonable when compared to the size of the smallest feature of the hub carrier before optimization (around 12 mm). A split draw constraint was used along y-direction to let the algorithm remove material from both sides (perpendicular to the y-axis) independently. It was decided to first try the “no through holes” option, being more manufacturable than the opposite option, find a reasonably low mass fraction and then optimize the component also allowing through holes to compare the results.

The following responses were added: mass fraction (design property only), weighted compliance based on the 6 load steps, rotation around x-axis and z-axis at the centre of the hub carrier.

The objective function was defined as the minimization of the weighted compliance, with unitary weights, so giving equal contribution to all the load steps.

The three constraints of the topology optimization problem were defined as follows:

- Mass fraction constraint, using the mass fraction response, was iteratively reduced starting from an upper bound of 0.5 and ending to the lowest value for which it's still possible to satisfy the constraints set on camber and toe stiffness.

- The two stiffness constraints were defined starting from value of the moments applied for stiffness evaluation and the minimum values of stiffness required

$$M_{x_{camberStiffEval}} = 160000 \text{ Nmm}$$

$$M_{z_{toeStiffEval}} = 84000 \text{ Nmm}$$

$$\phi_{x_{max}} = \frac{M_{x_{camberStiffEval}}}{MinCamberStiffness} = \frac{160000 \text{ Nmm}}{160000 * 1000 \text{ Nmm/rad}} = 0.001 \text{ rad}$$

$$\phi_{z_{max}} = \frac{M_{z_{toeStiffEval}}}{MinToeStiffness} = \frac{84000 \text{ Nmm}}{84000 * 1000 \text{ Nmm/rad}} = 0.001 \text{ rad}$$

So, for the stiffness constraints the upper bound of the rotation around the x and z axis at the centre of the hub was set to 0.001 rad.

Results of the topology optimization and structural analysis

Starting from 0.5 as upper bound of the mass fraction constraint, at first it was decreased down to 0.2 for both no-through-holes and through-holes option (manufacturing constraint), for which it was still possible to reach convergence. The final density frequency distribution of both options are compared in the following Figure 9.

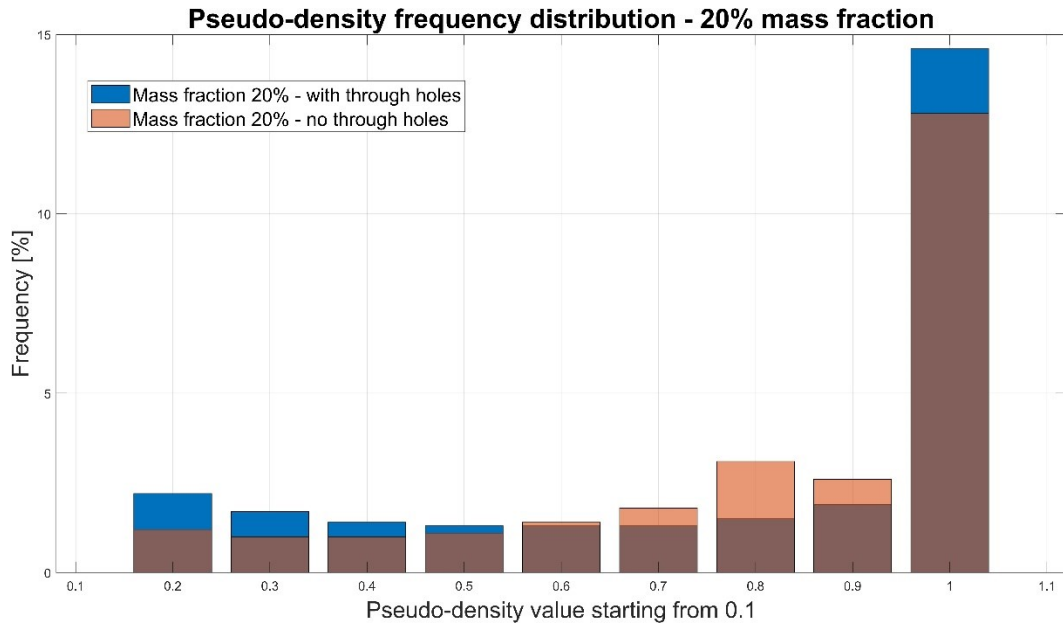


Figure 9: relative density frequency distribution for the optimized hub carrier (20% mass fraction), neglecting the 0-0.1 slot.

The trend of the weighted compliance of the optimization algorithm with 20% mass fraction for both through-holes and no-through-holes option is shown in Figure 10.

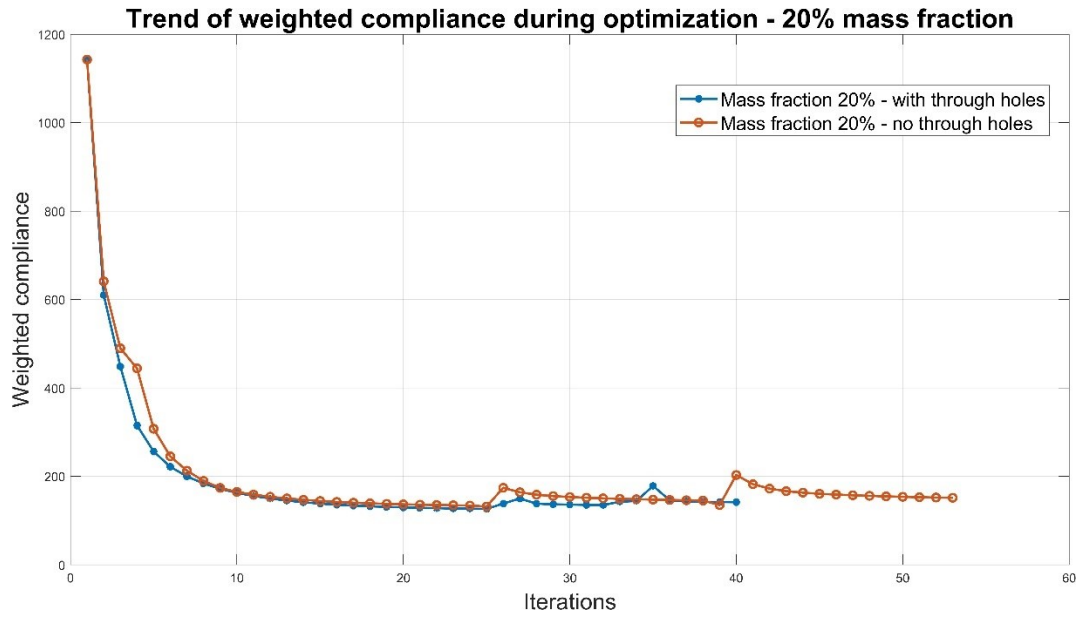


Figure 10: trend of the weighted compliance of the topology optimization algorithm for a 20% mass fraction constraint.

It was decided not to decrease even further the mass fraction of the with-through-holes component because to extract the optimized geometry, with its thin features, it was necessary to use a low density threshold, suggesting that a decrease of the mass fraction constraint would only offset the relative density frequency distribution towards the left (intermediate densities), without actually changing the mass of the post-processed component.

The initial design component had a mass of 1.972 kg, while the overall mass of the hub carrier was 2.089 kg. After the post-processing (pseudo-density thresholding and surface smoothing) of both the optimized components, their total mass is reported in the following Table 1.

	Total mass after post-processing	Actual mass fraction after post-processing (design component)
20% mass fraction – no through holes option	0.519 kg	20.4 %
20% mass fraction – with through holes option	0.559 kg	22.4 %

Table 1: summary of the total mass and mass fraction (design component only) after optimization.

After some FE analysis of the optimized hub carriers, the following results were obtained in terms of stiffness, recalling that 160000 Nm/rad and 84000 Nm/rad were the minimum camber and toe stiffness required respectively:

	Camber stiffness of the optimized component	Toe stiffness of the optimized component
20% mass fraction – no through holes option	522364 Nm/rad	121002 Nm/rad
20% mass fraction – with through holes option	597608 Nm/rad	136785 Nm/rad

Table 2: camber and toe stiffness obtained for the two optimized versions of the hub carrier, in green are highlighted the values satisfying the imposed behavioural constraint.

As the optimized component with the through-holes option is surely more difficult to manufacture than the other alternative, as Figure 11 shows, it was decided to discard this option and keep only the non-through holes option.

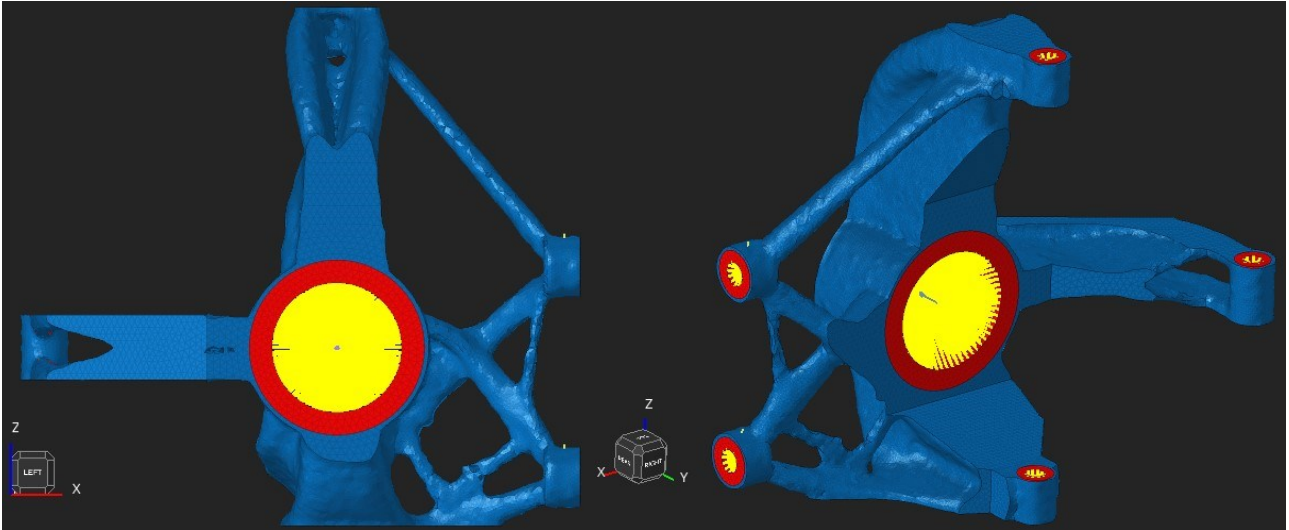


Figure 11: optimized hub carrier with 20% mass fraction and forging manufacturing constraint (holes allowed).

It was then decreased the mass fraction of the non-through-holes component, still reaching convergence down to 15% after 66 iterations, while the 10% mass fraction optimization never reached convergence in a reasonable number (80) of iterations even, if the constraints were satisfied. The trend of the weighted compliance is displayed in Figure 12 for both cases.

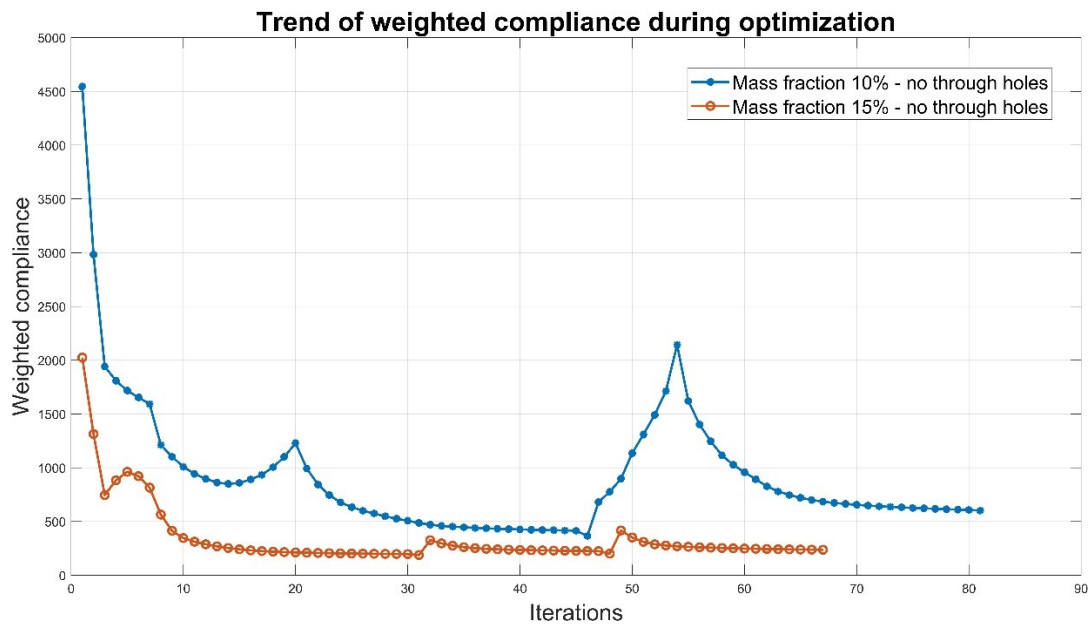


Figure 12: trend of the weighted compliance of the topology optimization algorithm for both 10% and 15% mass fraction constraint.

The relative density frequency distribution for the converged 15% mass fraction case is shown in Figure 13.

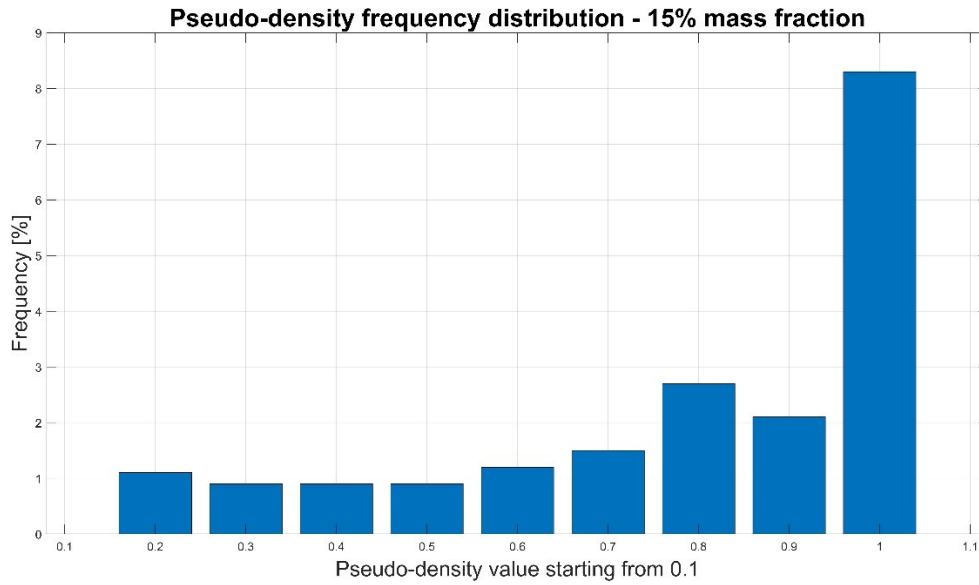


Figure 13: relative density frequency distribution for the optimized hub carrier (15% mass fraction), neglecting the 0-0.1 slot.

The optimized hub carrier obtained setting 15% mass fraction and split draw manufacturing constraint with “no through holes” option is shown in Figure 14.

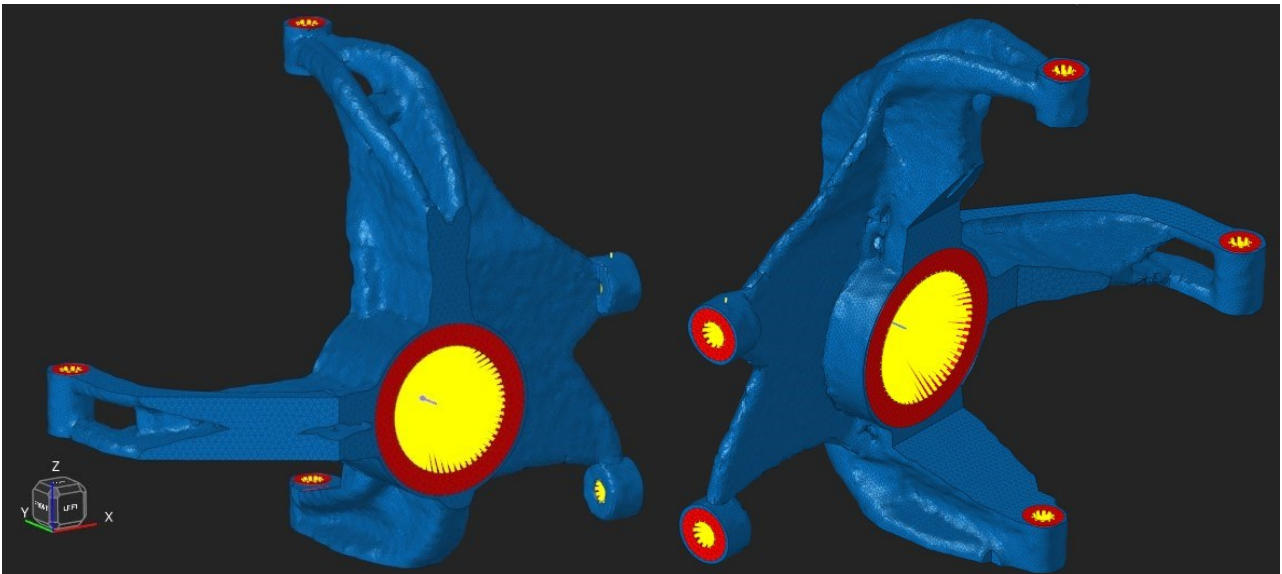


Figure 14: optimized hub carrier with 15% mass fraction and forging manufacturing constraint (holes not allowed).

As it is clear from the picture above, two through holes are found in this optimized version of the hub carrier despite the “no through holes” option being activated. Fortunately, these holes are simple to manufacture, so they don’t invalidate the results just obtained.

After the post-processing of the optimized component, to smooth the surfaces and avoid difficult-to-manufacture features, the following results were obtained in terms of mass reduction:

	Total mass after post-processing	Actual mass fraction after post-processing (design component)
20% mass fraction – no through holes option	0.460 kg	17.4 %

Table 3: total mass and mass fraction (design component only) after optimization with 15% mass fraction constraint.

While the camber and toe stiffness were computed to be:

	Camber stiffness of the optimized component	Toe stiffness of the optimized component
20% mass fraction – no through holes option	401617 Nm/rad	130511 Nm/rad

Table 4: camber and toe stiffness obtained for the optimized hub carrier, in green are highlighted the values satisfying the imposed behavioural constraint.

Finally, the results of the FE analysis in the most critical load case (internal obstacle) are represented in the figures below. These are not to take as definite, due to the poor mesh quality and missing mesh independence study, moreover, a safety factor computation wouldn't be representative.

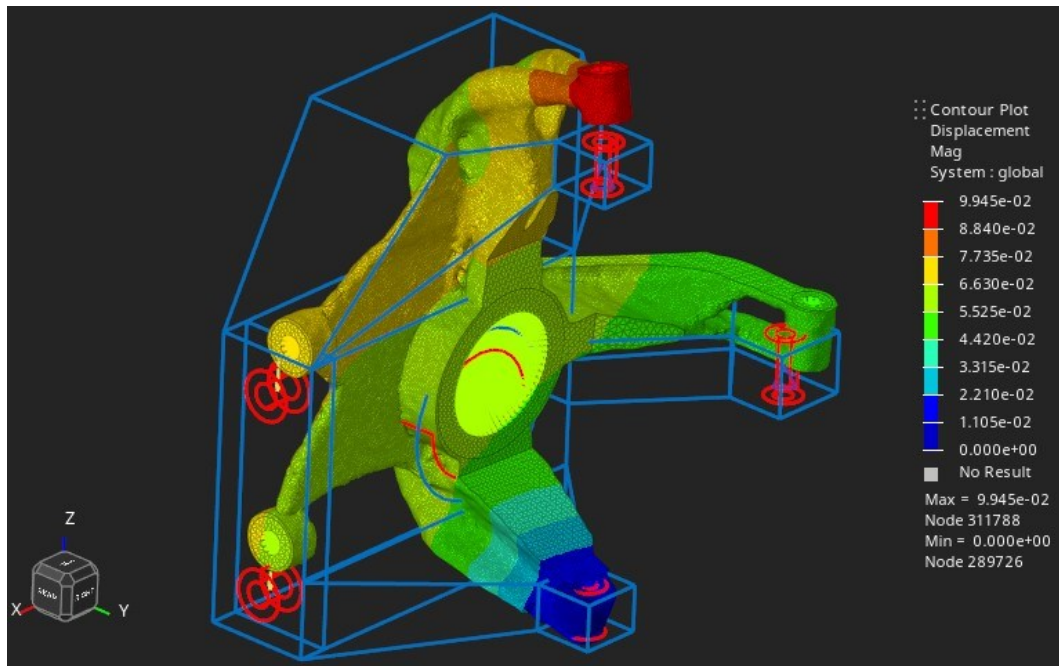


Figure 15: contour plot of the displacement magnitude field (deformation scale factor = 250).

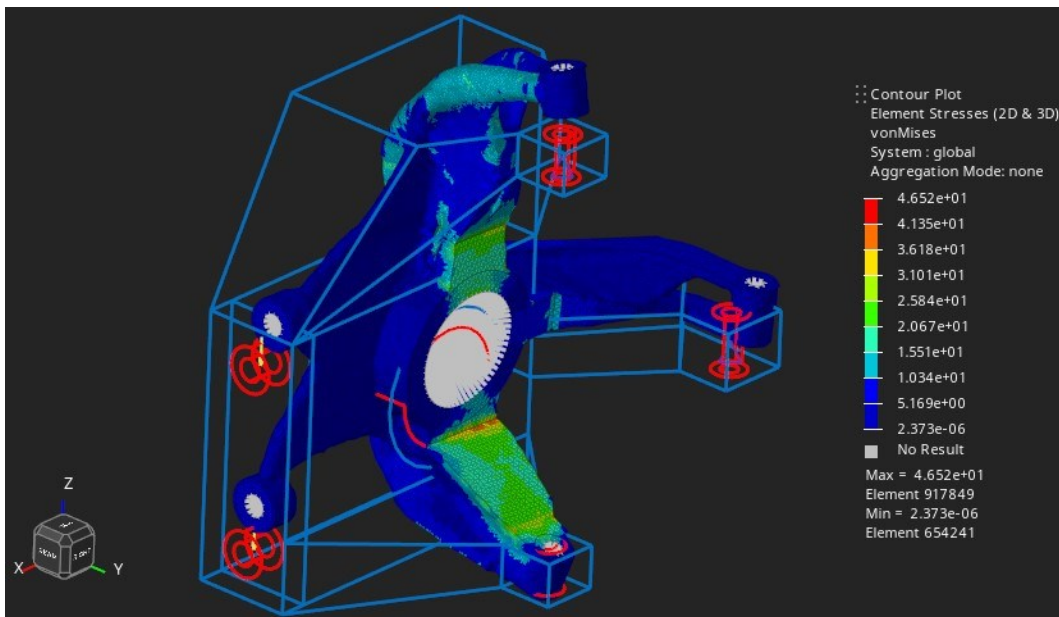


Figure 16: contour plot of the Von Mises stress field (deformation scale factor = 250).